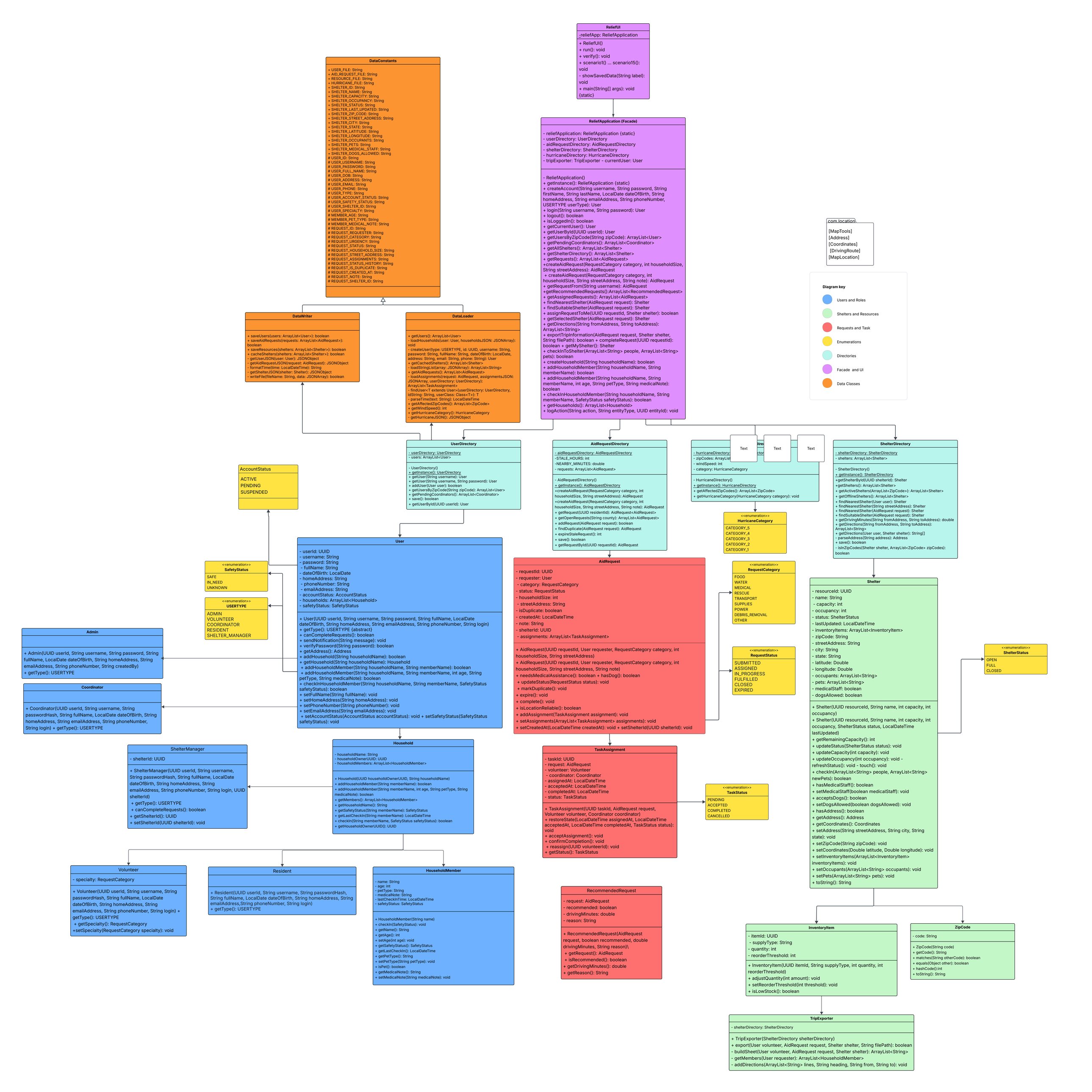

ReliefUI
-reliefApp: ReliefApplication
+ ReliefUI()
+ run(): void
+ verify(): void
+ scenario1() ... scenario15(): void
- showSavedData(String label): void
+ main(String[] args): void {static}
ReliefApplication (Facade)
- reliefApplication: ReliefApplication {static}
- userDirectory: UserDirectory
- aidRequestDirectory: AidRequestDirectory
- shelterDirectory: ShelterDirectory
- hurricaneDirectory: HurricaneDirectory
- tripExporter: TripExporter - currentUser: User
- ReliefApplication()
+ getInstance(): ReliefApplication {static}
+ createAccount(String username, String password, String firstName, String lastName, LocalDate dateOfBirth, String homeAddress, String emailAddress, String phoneNumber, USERTYPE userType): User
+ login(String username, String password): User
+ logout(): boolean
+ isLoggedIn(): boolean
+ getCurrentUser(): User
+ getUserById(UUID userId): User
+ getUsersByZipCode(String zipCode): ArrayList<User>
+ getPendingCoordinators(): ArrayList<Coordinator>
+ getAllShelters(): ArrayList<Shelter>
+ getShelterDirectory(): ArrayList<Shelter>
+ getRequests(): ArrayList<AidRequest>
+createAidRequest(RequestCategory category, int householdSize, String streetAddress): AidRequest
+ createAidRequest(RequestCategory category, int householdSize, String streetAddress, String note): AidRequest
+ getRequestFrom(String username): AidRequest
+getRecommendedRequests():ArrayList<RecommendedRequest>
+ getAssignedRequests(): ArrayList<AidRequest>
+ findNearestShelter(AidRequest request): Shelter
+ findSuitableShelter(AidRequest request): Shelter
+ assignRequestToMe(UUID requestId, Shelter shelter): boolean
+ getSelectedShelter(AidRequest request): Shelter
+ getDirections(String fromAddress, String toAddress): ArrayList<String>
+ exportTripInformation(AidRequest request, Shelter shelter, String filePath): boolean + completeRequest(UUID requestId): boolean + getMyShelter(): Shelter
+ checkInToShelter(ArrayList<String> people, ArrayList<String> pets): boolean
+ createHousehold(String householdName): boolean
+ addHouseholdMember(String householdName, String memberName): boolean
+ addHouseholdMember(String householdName, String memberName, int age, String petType, String medicalNote): boolean
+ checkInHouseholdMember(String householdName, String memberName, SafetyStatus safetyStatus): boolean
+ getHouseholds(): ArrayList<Household>
+ logAction(String action, String entityType, UUID entityId): void
DataConstants
+ USER_FILE: String
+ AID_REQUEST_FILE: String
+ RESOURCE_FILE: String
+ HURRICANE_FILE: String
+ SHELTER_ID: String
+ SHELTER_NAME: String
+ SHELTER_CAPACITY: String
+ SHELTER_OCCUPANCY: String
+ SHELTER_STATUS: String
+ SHELTER_LAST_UPDATED: String
+ SHELTER_ZIP_CODE: String
+ SHELTER_STREET_ADDRESS: String
+ SHELTER_CITY: String
+ SHELTER_STATE: String
+ SHELTER_LATITUDE: String
+ SHELTER_LONGITUDE: String
+ SHELTER_OCCUPANTS: String
+ SHELTER_PETS: String
+ SHELTER_MEDICAL_STAFF: String
+ SHELTER_DOGS_ALLOWED: String
# USER_ID: String
# USER_USERNAME: String
# USER_PASSWORD: String
# USER_FULL_NAME: String
# USER_DOB: String
# USER_ADDRESS: String
# USER_EMAIL: String
# USER_PHONE: String
# USER_TYPE: String
# USER_ACCOUNT_STATUS: String
# USER_SAFETY_STATUS: String
# USER_SHELTER_ID: String
# USER_SPECIALTY: String
# MEMBER_AGE: String
# MEMBER_PET_TYPE: String
# MEMBER_MEDICAL_NOTE: String
# REQUEST_ID: String
# REQUEST_REQUESTER: String
# REQUEST_CATEGORY: String
# REQUEST_URGENCY: String
# REQUEST_STATUS: String
# REQUEST_HOUSEHOLD_SIZE: String
# REQUEST_STREET_ADDRESS: String
# REQUEST_ASSIGNMENTS: String
# REQUEST_STATUS_HISTORY: String
# REQUEST_IS_DUPLICATE: String
# REQUEST_CREATED_AT: String
# REQUEST_NOTE: String
# REQUEST_SHELTER_ID: String
DataWriter
+ saveUsers(users: ArrayList<User>): boolean
+ saveAidRequests(requests: ArrayList<AidRequest>): boolean
+ saveResources(shelters: ArrayList<Shelter>): boolean
+ cacheShelters(shelters: ArrayList<Shelter>): boolean
- getUserJSON(user: User): JSONObject
- getAidRequestJSON(request: AidRequest): JSONObject
- formatTime(time: LocalDateTime): String
- getShelterJSON(shelter: Shelter): JSONObject
- writeFile(fileName: String, data: JSONArray): boolean
DataLoader
+ getUsers(): ArrayList<User>
- loadHouseholds(user: User, householdsJSON: JSONArray): void
- createUser(type: USERTYPE, id: UUID, username: String, password: String, fullName: String, dateOfBirth: LocalDate, address: String, email: String, phone: String): User
+ getCachedShelters(): ArrayList<Shelter>
- loadStringList(array: JSONArray): ArrayList<String>
+ getAidRequests(): ArrayList<AidRequest>
- loadAssignments(request: AidRequest, assignmentsJSON: JSONArray, userDirectory: UserDirectory): ArrayList<TaskAssignment>
- findUser<T extends User>(userDirectory: UserDirectory, idString: String, userClass: Class<T>): T
- parseTime(text: String): LocalDateTime
+ getAffectedZipCodes(): ArrayList<ZipCode>
+ getWindSpeed(): int
+ getHurricaneCategory(): HurricaneCategory
- getHurricaneJSON(): JSONObject
com.location
[MapTools]
[Address]
[Coordinates]
[DrivingRoute]
[MapLocation]
Diagram key
Users and Roles
Shelters and Resources
Requests and Task
Enumerations
Directories
Facade and UI
Data Classes
AccountStatus
ACTIVE
PENDING
SUSPENDED
UserDirectory
- userDirectory: UserDirectory
- users: ArrayList<User>
- UserDirectory()
+ getInstance(): UserDirectory
+ getUser(String username): User
+ getUser(String username, String password): User
+ addUser(User user): boolean
+ getUsersByZipCode(String zipCode): ArrayList<User>
+ getPendingCoordinators(): ArrayList<Coordinator>
+ save(): boolean
+ getUserById(UUID userId): User
AidRequestDirectory
- aidRequestDirectory: AidRequestDirectory
-STALE_HOURS: int
-NEARBY_MINUTES: double
- requests: ArrayList<AidRequest>
- AidRequestDirectory()
+ getInstance(): AidRequestDirectory
+createAidRequest(RequestCategory category, int householdSize, String streetAddress): AidRequest
+createAidRequest(RequestCategory category, int householdSize, String streetAddress, String note): AidRequest
+ getRequest(UUID residentId): AidRequest<AidRequest>
+ getOpenRequests(String county): ArrayList<AidRequest>
+ addRequest(AidRequest request): boolean
+ findDuplicate(AidRequest request): AidRequest
+ expireStaleRequest(): int
+ save(): boolean
+ getRequestById(UUID requestId): AidRequest
Text
Text
Text
- hurricaneDirectory:
- zipCodes: ArrayList
- windSpeed: int
- category: HurricaneCategory
- HurricaneDirectory()
+ getInstance(): HurricaneDirectory
+ getAffectedZipCodes(): ArrayList<ZipCode>
+ getHurricaneCategory(HurricaneCategory category): void
<<enumeration>>
HurricaneCategory
CATEGORY_5
CATEGORY_4
CATEGORY_3
CATEGORY_2
CATEGORY_1
ShelterDirectory
- shelterDirectory: ShelterDirectory
- shelters: ArrayList<Shelter>
- ShelterDirectory()
+ getInstance(): ShelterDirectory
+getShelterById(UUID shelterId): Shelter
+getShelters(): ArrayList<Shelter>
+ getActiveShelters(ArrayList<ZipCode>): ArrayList<Shelter>
+ getOfflineShelters(): ArrayList<Shelter>
+ findNearestShelter(User user): Shelter
+ findNearestShelter(String streetAddress): Shelter
+ findNearestShelter(AidRequest request): Shelter
+ findSuitableShelter(AidRequest request): Shelter
+ getDrivingMinutes(String fromAddress, String toAddress): double
+ getDirections(String fromAddress, String toAddress): ArrayList<String>
+ getDirections(User user, Shelter shelter): String[]
- parseAddress(String address): Address
+ save(): boolean
- isInZipCodes(Shelter shelter, ArrayList<ZipCode> zipCodes): boolean
User
- userId: UUID
- username: String
- password: String
- fullName: String
- dateOfBirth: LocalDate
- homeAddress: String
- phoneNumber: String
- emailAddress: String
- accountStatus: AccountStatus
- households: ArrayList<Household>
- safetyStatus: SafetyStatus
+ User(UUID userId, String username, String password, String fullName, LocalDate dateOfBirth, String homeAddress, String emailAddress, String phoneNumber, String login)
+ getType(): USERTYPE {abstract}
+ canCompleteRequests(): boolean
+ sendNotification(String message): void
+ verifyPassword(String password): boolean
+ getAddress(): Address
+ addHousehold(String householdName): boolean
+ getHousehold(String householdName): Household
+ addHouseholdMember(String householdName, String memberName): boolean
+ addHouseholdMember(String householdName, String memberName, int age, String petType, String medicalNote): boolean
+ checkInHouseholdMember(String householdName, String memberName, SafetyStatus safetyStatus): boolean
+ setFullName(String fullName): void
+ setHomeAddress(String homeAddress): void
+ setPhoneNumber(String phoneNumber): void
+ setEmailAddress(String emailAddress): void
+ setAccountStatus(AccountStatus accountStatus): void + setSafetyStatus(SafetyStatus safetyStatus): void
<<enumeration>>
SafetyStatus
SAFE
IN_NEED
UNKNOWN
<<enumeration>>
USERTYPE
ADMIN
VOLUNTEER
COORDINATOR
RESIDENT
SHELTER_MANAGER
AidRequest
- requestId: UUID
- requester: User
- category: RequestCategory
- status: RequestStatus
- householdSize: int
- streetAddress: String
- isDuplicate: boolean
- createdAt: LocalDateTime
- note: String
- shelterId: UUID
- assignments: ArrayList<TaskAssignment>
+ AidRequest(UUID requestId, User requester, RequestCategory category, int householdSize, String streetAddress)
+ AidRequest(UUID requestId, User requester, RequestCategory category, int householdSize, String streetAddress, String note)
+ needsMedicalAssistance(): boolean + hasDog(): boolean
+ updateStatus(RequestStatus status): void
+ markDuplicate(): void
+ expire(): void
+ complete(): void
+ isLocationReliable(): boolean
+ addAssignment(TaskAssignment assignment): void
+ setAssignments(ArrayList<TaskAssignment> assignments): void
+ setCreatedAt(LocalDateTime createdAt): void + setShelterId(UUID shelterId): void
<<enumeration>>
RequestCategory
FOOD
WATER
MEDICAL
RESCUE
TRANSPORT
SUPPLIES
POWER
DEBRIS_REMOVAL
OTHER
<<enumeration>>
RequestStatus
SUBMITTED
ASSIGNED
IN_PROGRESS
FULFILLED
CLOSED
EXPIRED
Shelter
- resourceId: UUID
- name: String
- capacity: int
- occupancy: int
- status: ShelterStatus
- lastUpdated: LocalDateTime
- inventoryItems: ArrayList<InventoryItem>
- zipCode: String
- streetAddress: String
- city: String
- state: String
- latitude: Double
- longitude: Double
- occupants: ArrayList<String>
- pets: ArrayList<String>
- medicalStaff: boolean
- dogsAllowed: boolean
+ Shelter(UUID resourceId, String name, int capacity, int occupancy)
+ Shelter(UUID resourceId, String name, int capacity, int occupancy, ShelterStatus status, LocalDateTime lastUpdated)
+ getRemainingCapacity(): int
+ updateStatus(ShelterStatus status): void
+ updateCapacity(int capacity): void
+ updateOccupancy(int occupancy): void - refreshStatus(): void - touch(): void
+ checkIn(ArrayList<String> people, ArrayList<String> newPets): boolean
+ hasMedicalStaff(): boolean
+ setMedicalStaff(boolean medicalStaff): void
+ acceptsDogs(): boolean
+ setDogsAllowed(boolean dogsAllowed): void
+ hasAddress(): boolean
+ getAddress(): Address
+ getCoordinates(): Coordinates
+ setAddress(String streetAddress, String city, String state): void
+ setZipCode(String zipCode): void
+ setCoordinates(Double latitude, Double longitude): void
+ setInventoryItems(ArrayList<InventoryItem> inventoryItems): void
+ setOccupants(ArrayList<String> occupants): void
+ setPets(ArrayList<String> pets): void
+ toString(): String
<<enumeration>>
ShelterStatus
OPEN
FULL
CLOSED
Admin
+ Admin(UUID userId, String username, String password, String fullName, LocalDate dateOfBirth, String homeAddress, String emailAddress, String phoneNumber, String createdBy)
+ getType(): USERTYPE
Coordinator
+ Coordinator(UUID userId, String username, String passwordHash, String fullName, LocalDate dateOfBirth, String homeAddress, String emailAddress, String phoneNumber, String login) + getType(): USERTYPE
ShelterManager
- shelterId: UUID
+ ShelterManager(UUID userId, String username, String passwordHash, String fullName, LocalDate dateOfBirth, String homeAddress, String emailAddress, String phoneNumber, String login, UUID shelterId)
+ getType(): USERTYPE
+ canCompleteRequests(): boolean
+ getShelterId(): UUID
+ setShelterId(UUID shelterId): void
Household
- householdName: String
- householdOwnerUUID: UUID
- householdMembers: ArrayList<HouseholdMember>
+ Household(UUID householdOwnerUUID, String householdName)
+ addHouseholdMember(String memberName): boolean
+ addHouseholdMember(String memberName, int age, String petType, String medicalNote): boolean
+ getMembers(): ArrayList<HouseholdMember>
+ getHouseholdName(): String
+ getSafetyStatus(String memberName): SafetyStatus
+ getLastCheckIn(String memberName): LocalDateTime
+ checkIn(String memberName, SafetyStatus safetyStatus): boolean
+ getHouseholdOwnerUUID(): UUID
TaskAssignment
- taskId: UUID
- request: AidRequest
- volunteer: Volunteer
- coordinator: Coordinator
- assignedAt: LocalDateTime
- acceptedAt: LocalDateTime
- completedAt: LocalDateTime
- status: TaskStatus
+ TaskAssignment(UUID taskId, AidRequest request, Volunteer volunteer, Coordinator coordinator)
+ restoreState(LocalDateTime assignedAt, LocalDateTime acceptedAt, LocalDateTime completedAt, TaskStatus status): void
+ acceptAssignment(): void
+ confirmCompletion(): void
+ reassign(UUID volunteerId): void
+ getStatus(): TaskStatus
<<enumeration>>
TaskStatus
PENDING
ACCEPTED
COMPLETED
CANCELLED
Volunteer
- specialty: RequestCategory
+ Volunteer(UUID userId, String username, String passwordHash, String fullName, LocalDate dateOfBirth, String homeAddress, String emailAddress, String phoneNumber, String login) + getType(): USERTYPE
+ getSpecialty(): RequestCategory
+setSpecialty(RequestCategory specialty): void
Resident
+ Resident(UUID userId, String username, String passwordHash, String fullName, LocalDate dateOfBirth, String homeAddress, String emailAddress,String phoneNumber, String login)
+ getType(): USERTYPE
HouseholdMember
- name: String
- age: int
- petType: String
- medicalNote: String
- lastCheckInTime: LocalDateTime
- safetyStatus: SafetyStatus
+ HouseholdMember(String name)
+ checkIn(SafetyStatus): void
+ getName(): String
+ getAge(): int
+ setAge(int age): void
+ getSafetyStatus(): SafetyStatus
+ getLastCheckIn(): LocalDateTime
+ getPetType(): String
+ setPetType(String petType): void
+ isPet(): boolean
+ getMedicalNote(): String
+ setMedicalNote(String medicalNote): void
RecommendedRequest
- request: AidRequest
- recommended: boolean
- drivingMinutes: double
- reason: String
+ RecommendedRequest(AidRequest request, boolean recommended, double drivingMinutes, String reason)\
+ getRequest(): AidRequest
+ isRecommended(): boolean
+ getDrivingMinutes(): double
+ getReason(): String
InventoryItem
- itemId: UUID
- supplyType: String
- quantity: int
- reorderThreshold: int
+ InventoryItem(UUID itemId, String supplyType, int quantity, int reorderThreshold)
+ adjustQuantity(int amount): void
+ setReorderThreshold(int threshold): void
+ isLowStock(): boolean
ZipCode
- code: String
+ ZipCode(String code)
+ getCode(): String
+ matches(String otherCode): boolean
+ equals(Object other): boolean
+ hashCode():int
+ toString(): String
TripExporter
- shelterDirectory: ShelterDirectory
+ TripExporter(ShelterDirectory shelterDirectory)
+ export(User volunteer, AidRequest request, Shelter shelter, String filePath): boolean
- buildSheet(User volunteer, AidRequest request, Shelter shelter): ArrayList<String>
- getMembers(User requester): ArrayList<HouseholdMember>
- addDirections(ArrayList<String> lines, String heading, String from, String to): void